

8.C-ARM MACHINE

TECHNICAL SPECIFICATION

1. GENERAL REQUIREMENTS

Sl. No.	Parameter	Minimum Tender Requirement
1	Equipment	Mobile digital C-Arm X-ray imaging system
2	Application	Orthopaedics, Trauma, Spine, Urology, General Surgery, Pain Management and other intraoperative procedures
3	Generator	High-frequency X-ray generator
4	Maximum Generator Power	5- 15 kW
5	Generator Frequency	Minimum 50 kHz or higher
6	X-ray Tube	Rotating anode or stationary anode, as per system configuration
7	Focal Spot	Small focal spot suitable for high-resolution fluoroscopy
8	kV Range	Approximately 40–120 kV or wider
9	mA Range	Minimum suitable range for fluoroscopy and radiography
10	Fluoroscopy	Continuous and pulsed fluoroscopy
11	Digital Imaging	Real-time digital image acquisition and processing
12	Image Display	High-resolution medical-grade LCD/LED monitors
13	Image Intensifier / FPD	9"/12" or larger image intensifier OR flat-panel detector of equivalent/superior performance
14	Digital Resolution	Minimum 1K × 1K image matrix or better
15	Image Storage	Minimum 10,000 images or equivalent digital storage
16	DICOM	DICOM 3.0 compatibility
17	Image Processing	Real-time digital image processing
18	Collimation	Motorized or software-controlled collimation
19	Dose Reduction	Pulsed fluoroscopy and dose-saving features
20	C-Arm Movement	Motorized or manually assisted C-arm positioning
21	Orbital Movement	Minimum approximately ±90° or better
22	Horizontal Movement	Minimum approximately 20 cm or better
23	Vertical Movement	Minimum approximately 40 cm or better
24	C-Arm Depth	Minimum approximately 65 cm or better
25	C-Arm Rotation	Minimum approximately ±180° or equivalent
26	SID	Suitable for intraoperative imaging
27	Sterile Operation	Suitable for use in sterile OT environment

Sl. No.	Parameter	Minimum Tender Requirement
28	Mobile Design	Compact mobile trolley with locking wheels
29	Power Supply	220–240 V AC, 50 Hz
30	Emergency	Emergency stop and system safety features

2. X-RAY GENERATOR

The system shall have:

- High-frequency X-ray generator.
- Maximum output power: **≥15 kW**.
- Generator frequency: **≥50 kHz**.
- Microprocessor-controlled generator.
- Automatic exposure control, where applicable.
- Fluoroscopy and radiography modes.
- Continuous fluoroscopy.
- Pulsed fluoroscopy.
- Low-dose fluoroscopy.
- High-dose mode for difficult anatomical regions, where applicable.
- Automatic brightness/dose rate control.
- kV and mA adjustment.
- Overload protection.
- Tube protection circuitry.
- Automatic exposure termination.

3. X-RAY TUBE

The X-ray tube shall have:

- High-performance tube suitable for prolonged fluoroscopic procedures.
- Small focal spot for high-resolution imaging.
- Adequate heat storage capacity.
- Tube cooling mechanism.
- Tube overload protection.
- Tube temperature monitoring.
- Automatic tube protection.
- Suitable anode design for continuous intraoperative applications.

Recommended tender clause:

"The X-ray tube shall be capable of prolonged fluoroscopic operation and shall have adequate heat storage and cooling capacity for continuous intraoperative procedures. The bidder shall

clearly specify anode heat storage capacity, focal spot size, maximum tube current and tube cooling characteristics."

4. FLAT PANEL DETECTOR

The system shall be offered with:

Flat Panel Detector

- Digital flat-panel detector.
- Minimum detector size suitable for orthopaedic and general surgical applications.
- High DQE.
- High spatial resolution.
- Low image lag.
- High dynamic range.
- Suitable for pulsed fluoroscopy.
- Easy-to-clean surface.
- Detector size 30 cm x 30 cm

5. DIGITAL IMAGING SYSTEM

The system shall provide:

- Real-time digital image acquisition.
- Minimum **1024 × 1024 (1K × 1K)** image matrix or better.
- High-resolution image display.
- Last Image Hold (LIH).
- Image freeze.
- Image reversal.
- Image rotation.
- Image zoom.
- Image annotation.
- Measurement tools.
- Image enhancement.
- Edge enhancement.
- Noise reduction.
- Contrast and brightness adjustment.
- Image subtraction, where applicable.
- Multi-image display.
- Patient demographic entry.
- Image archival.
- USB/network data export.
- DICOM connectivity.

6. MONITORS

The system shall be supplied with:

- Minimum **2 medical-grade high-resolution monitors**.
- Minimum **19-inch or larger LCD/LED monitors** preferred.
- One monitor for live fluoroscopy.
- One monitor for reference/previous images.
- Independent monitor adjustment.
- High brightness and contrast.
- Anti-glare display.
- Suitable viewing angle.
- Medical-grade display with appropriate electrical safety.

Optional: Third monitor for technician/operator viewing may be considered.

7. MECHANICAL SPECIFICATIONS

The C-arm shall provide:

- Easy positioning around the patient.
- Motorized or assisted vertical movement.
- Orbital movement.
- Horizontal movement.
- C-arm rotation.
- Wig-wag movement, where applicable.
- Parking position.
- Electromagnetic or mechanical brakes.
- Large sterile field access.
- Easy movement around OT table.
- Compact footprint.
- Lockable wheels.
- Robust construction.

Recommended minimum mechanical parameters

Parameter	Minimum
Vertical travel	≥ 400 mm
Orbital movement	$\geq \pm 90^\circ$
Horizontal movement	≥ 200 mm
C-arm rotation	$\geq \pm 180^\circ$
C-arm depth	≥ 650 mm
SID	≥ 900 mm preferred

The bidder shall submit exact mechanical movement specifications for the offered model.

8. FLUOROSCOPY & DOSE MANAGEMENT

The system shall include:

- Continuous fluoroscopy.
- Pulsed fluoroscopy.
- Low-dose fluoroscopy.
- Automatic dose rate control.
- Automatic brightness control.
- Last Image Hold.
- Fluoroscopy time display.
- Dose monitoring.
- Dose-area product (DAP) display, where available.
- Digital image storage.
- Collimation without unnecessary exposure.
- Radiation dose reduction algorithms.

9. COLLIMATION

The system shall have:

- Adjustable X-ray collimation.
- Iris collimator or equivalent.
- Rectangular shutters.
- Software-controlled collimation where applicable.
- Collimation without additional radiation exposure, preferably.
- Easy operator control.

10. SOFTWARE

The system shall provide:

- Patient registration.
- Patient data management.
- Image acquisition.
- Image review.
- Image processing.
- Image enhancement.
- Image annotation.
- Image measurement.

- Image rotation and flipping.
- Image zoom.
- Image storage.
- Image retrieval.
- DICOM 3.0.
- DICOM Storage.
- DICOM Print.
- DICOM Worklist.
- DICOM Query/Retrieve.
- Network connectivity.
- USB export.
- PACS compatibility.

11. APPLICATIONS

The C-arm shall be suitable for:

Orthopaedics

- Trauma surgery
- Fracture fixation
- Intramedullary nailing
- Plating
- Screw fixation
- Spine surgery
- Pedicle screw placement
- Joint procedures

Urology

- PCNL
- DJ stent placement
- Ureteric procedures
- Stone procedures

General Surgery

- Intraoperative imaging
- Foreign body localization
- Drainage procedures

Pain Management

- Spine injections
- Nerve blocks
- Fluoroscopy-guided interventions

Other

- Vascular procedures, subject to system configuration
- Gastrointestinal procedures
- Emergency procedures

12. RADIATION SAFETY

The system shall comply with applicable:

- AERB requirements in India.
- Applicable IEC standards for medical electrical equipment.
- Applicable radiation safety regulations.
- Manufacturer's radiation safety requirements.

The supplier shall provide:

- AERB type approval/registration documentation, as applicable.
- Radiation safety documentation.
- Installation report.
- Quality assurance report.
- Radiation leakage test report.
- Calibration documentation.

The bidder shall be responsible for assisting the purchaser with applicable **AERB regulatory compliance** before clinical use.

13. REQUIRED ACCESSORIES

The system shall be supplied with:

1. C-arm – **1 No.**
2. X-ray tube – **1 No.**
3. Image intensifier/FPD – **1 No.**
4. Medical-grade monitor – **minimum 2 Nos.**
5. Control console – **1 No.**
6. Footswitch – **1 No.**
7. DICOM interface – **1 No.**
8. Patient table interface/accessories – as required
9. Radiation protection accessories
10. Power cable
11. User manual
12. Service manual
13. Installation and commissioning

14. Application training
15. Biomedical engineer training
16. DICOM/PACS integration
17. Radiation safety documentation

14. WARRANTY

Parameter	Minimum Requirement
Comprehensive Warranty	Minimum 3 years
X-ray Tube Warranty	Minimum 5 years or as offered
Detector/Image Intensifier	Minimum 5 years
Generator	Minimum 5 years
Service Response	Within 24 hours
Preventive Maintenance	Minimum 6 visits/year during warranty
Spare Parts	Minimum 7–10 years availability
Service Support	Authorized service support in India
Calibration	Periodic calibration as per manufacturer
Training	Radiologist/Surgeon, OT staff and Biomedical Engineer
Installation	Complete installation and commissioning
PACS Integration	Included where specified
AERB Assistance	Included